Description of the PV-RT Power System Cyberattack Benchmark Model

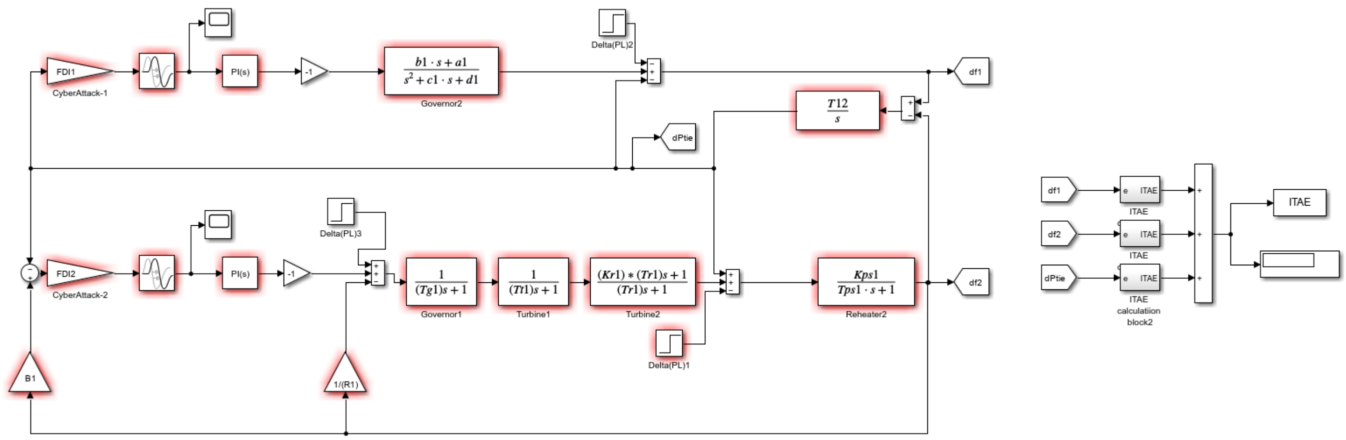

Figure 1. Simulink model of the PV-RT power system cyberattack benchmark system.

This Simulink model was developed to investigate the behavior of a PV-integrated two-area load frequency control (LFC) system under cyberattack conditions and to compare different controller structures in a fair and reproducible manner. The benchmark structure allows the same load disturbance and cyberattack scenarios to be applied to each tested controller. Therefore, the obtained performance values can be directly compared under identical operating and attack conditions.

The studied PV-RT power system consists of two interconnected control areas. The first area includes a photovoltaic (PV) power system, while the second area represents a reheat thermal power system. The two areas are connected through the tie-line power deviation dynamics, and the main dynamic outputs of interest are the frequency deviation in Area 1, the frequency deviation in Area 2, and the tie-line power deviation.

Deviation-Based Representation of the LFC System

In this benchmark model, the LFC system is represented using deviation variables around the nominal operating point. In normal operation, the scheduled frequency and scheduled tie-line power exchange are considered as the reference operating condition. Therefore, the model does not directly analyze the absolute frequency values, but rather the deviations from the nominal operating point.

For this reason, the desired values of the frequency deviations and tie-line power deviation are considered to be zero under steady-state nominal operation:

$$\Delta f_1 = 0, \quad \Delta f_2 = 0, \quad \Delta P_{tie} = 0$$

This expression indicates that the system is initially assumed to operate at the nominal frequency and scheduled tie-line power exchange. When a load disturbance or a cyberattack is applied, deviations occur in the system variables. The controller then attempts to suppress these deviations and restore the system to the nominal operating condition.

The output signals monitored in the model are Δf_1, Δf_2, and ΔP_{tie}. These signals are used both for visual analysis of the transient response and for the calculation of the ITAE-based performance index.

Main Blocks of the PV-RT Benchmark Model

The PV-side area is represented by a transfer-function-based dynamic model of the photovoltaic power system. This block reflects the dynamic behavior of the PV-integrated area in response to load and control actions. The reheat thermal area consists of governor, turbine, reheater, and power system blocks. These components model the conventional thermal generation dynamics that participate in frequency regulation.

The tie-line block represents the dynamic interaction between the two areas. The tie-line power deviation is obtained according to the frequency deviations of the interconnected areas. As a result, the model can evaluate not only the local frequency response of each area but also the inter-area power exchange response.

$$\Delta P_{tie} = (T_{12}/\, s)(\Delta f_1 - \Delta f_2)$$

The controller inputs are formed using the area control error (ACE) signals. In the PV-integrated area, the ACE signal is associated with the tie-line power deviation. In the reheat thermal area, the ACE signal is calculated using both the frequency deviation and the tie-line power deviation. In this way, the control system considers both local frequency regulation and inter-area power balancing.

$$ACE_1 = \Delta P_{tie}, \quad ACE_2 = B * \Delta f_2 - \Delta P_{tie}$$

Cyberattack and Load Disturbance Representation

The benchmark model includes load disturbance and cyberattack inputs so that controller robustness can be evaluated under realistic adverse conditions. The load disturbance signal represents random varying load conditions applied to the system. In the benchmark scenario file, this disturbance is represented by the SLP2 variable, which corresponds to the load change applied to the second area.

False data injection (FDI) and denial-of-service (DoS) attacks are applied to the ACE signals entering the controllers. The FDI attack distorts the ACE signal by multiplying it with an attack coefficient. In the benchmark model, FDI1 and FDI2 represent the FDI attack coefficients applied to Area 1 and Area 2, respectively. When the FDI coefficient is equal to 1, no FDI attack is applied. Values greater than 1 indicate that the ACE signal is proportionally manipulated.

The DoS attack is represented as a communication delay in the ACE signal path. In the benchmark model, DoS1 and DoS2 represent the delay values applied to Area 1 and Area 2, respectively. These delay values emulate the effect of a denial-of-service attack that prevents timely delivery of the control signal information to the controller.

$$ACE_1FDI(t) = ACE_1(t)xFDI_1, \quad ACE_2FDI(t) = ACE_2(t)xFDI_2$$
$$ACE_1DoS(t) = ACE_1(t - DoS_1), \quad ACE_2DoS(t) = ACE_2(t - DoS_2)$$

By changing these scenario variables, the benchmark model can test the controller under normal operation, FDI attacks, DoS attacks, and combined attack conditions. Since every controller is tested using the same scenario values, the comparison is fair and reproducible.

Controller Replacement for Benchmark Testing

The controller blocks in the Simulink model can be replaced by any controller structure proposed by a researcher. For example, the provided benchmark model may include PI, PPI, or FOPPI controller blocks depending on the test configuration. To evaluate a new controller, the researcher only needs to replace the existing controller block with the new controller structure and define the required controller parameters in the MATLAB workspace or in a separate parameter file.

After the controller block is replaced, the same benchmark scenario file can be used to run all cyberattack and load disturbance cases. The obtained ITAE values can then be compared with the reference controller results, such as the proposed FOPPI controller or the classical PPI controller results.

ITAE-Based Performance Evaluation

The ITAE criterion is used as the main performance index in the benchmark model. In the PV-RT LFC system, the control objective is not based on a single signal. Instead, the frequency deviations of both areas and the tie-line power deviation are considered together. Therefore, the ITAE value represents the accumulated time-weighted effect of Δf_1, Δf_2, and ΔP_{tie} throughout the simulation.

ITAE = integral from 0 to tsim of t ($|\Delta f_1| + |\Delta f_2| + |\Delta P_{tie}|$*) dt*

Since the ITAE output in the Simulink model is obtained as a cumulative signal calculated over time, the final value of this signal at the end of the simulation is used as the scenario-level performance value. In the MATLAB scripts, this value is obtained by using max(ITAE). Because the ITAE signal is cumulative, this maximum value corresponds to the final accumulated performance value at the end of the simulation.

A lower final ITAE value indicates that the controller suppresses the frequency deviations and tie-line power deviation more quickly and more effectively under load disturbance and cyberattack conditions. Therefore, the final ITAE values obtained for different scenarios are used to compare the robustness performance of different controllers.

Main Scenario Variables and Output Signals

Variable	Description
ID	Scenario number in the benchmark file.
SLP_2	Load disturbance or load change applied to Area 2.
FDI_1	False data injection coefficient applied to the Area 1 ACE signal.
FDI_2	False data injection coefficient applied to the Area 2 ACE signal.
DoS_1	Communication delay applied to the Area 1 ACE signal under DoS attack.
DoS_2	Communication delay applied to the Area 2 ACE signal under DoS attack.
Δf_1	Frequency deviation of Area 1.
Δf_2	Frequency deviation of Area 2.
ΔP_{tie}	Tie-line power deviation between the two areas.
ITAE	Time-weighted cumulative performance index used for controller comparison.

Purpose of the Benchmark Model

The main purpose of this model is to provide a reproducible test environment for analyzing cyberattack effects in PV-integrated LFC systems. The model enables a comparative investigation of both the effects of FDI and DoS attacks on the system response and the ability of different controllers to mitigate these effects.

With this benchmark structure, newly proposed or existing controllers can be tested under identical random load and cyberattack scenarios. Therefore, the robustness of different controllers in PV-integrated two-area LFC systems can be evaluated using the same reference conditions and the same ITAE-based performance criterion.